

Introduction to linear models

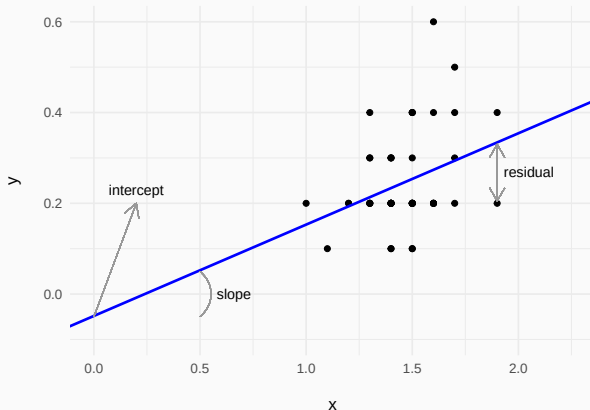
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Our unified regression framework (GLM)

$$y_i = a + bx_i + \varepsilon_i$$

$$\varepsilon_i \sim N(0, \sigma^2)$$



Data

y = response variable

x = predictor

Parameters

a = intercept

b = slope

σ = residual variation

ε = residuals

What's the intercept?

Expected value of y when predictors (x) = 0

If $x = 0$:

- $y = a + b \cdot 0$

What's the intercept?

Expected value of y when predictors (x) = 0

If $x = 0$:

- $y = a + b \cdot 0$
- $y = a$

What's the slope?

How much **y** increases (or decreases) when **x** increases in 1 unit

If we have model

$$y = 0.5 + 2 \cdot x$$

If **x** increases 1 unit, **y** increases **2 units**

- If $x = 10 \rightarrow y = 0.5 + 2 * 10 = 20.5$

What's the slope?

How much **y** increases (or decreases) when **x** increases in 1 unit

If we have model

$$y = 0.5 + 2 \cdot x$$

If **x** increases 1 unit, **y** increases **2 units**

- If $x = 10 \rightarrow y = 0.5 + 2 * 10 = 20.5$
- If $x = 11 \rightarrow y = 0.5 + 2 * 11 = 22.5$

Slopes can be negative

If we have model

$$y = 0.5 - 2x$$

If x increases 1 unit, y decreases 2 units

- If $x = 10 \rightarrow y = 0.5 - 2 * 10 = -19.5$

Slopes can be negative

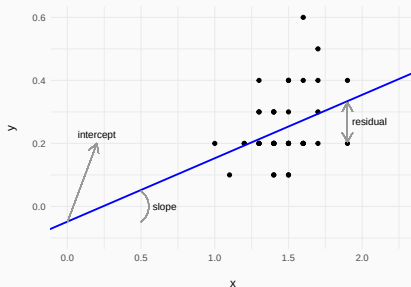
If we have model

$$y = 0.5 - 2x$$

If x increases 1 unit, y decreases 2 units

- If $x = 10 \rightarrow y = 0.5 - 2 * 10 = -19.5$
- If $x = 11 \rightarrow y = 0.5 - 2 * 11 = -21.5$

What are residuals?



How far points fall from the regression line

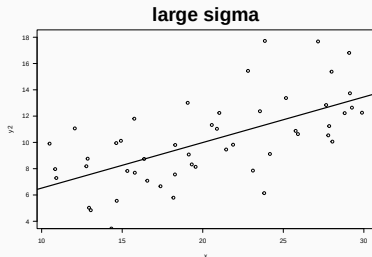
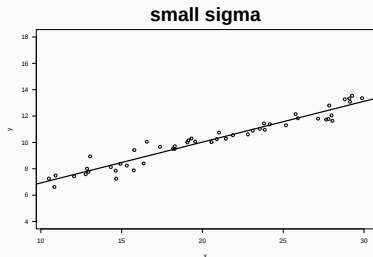
Difference between **observed values** and values **predicted** by model (regression line)

If sigma is large, residuals are larger

$$\varepsilon_i \sim N(0, \sigma^2)$$

If sigma is larger:

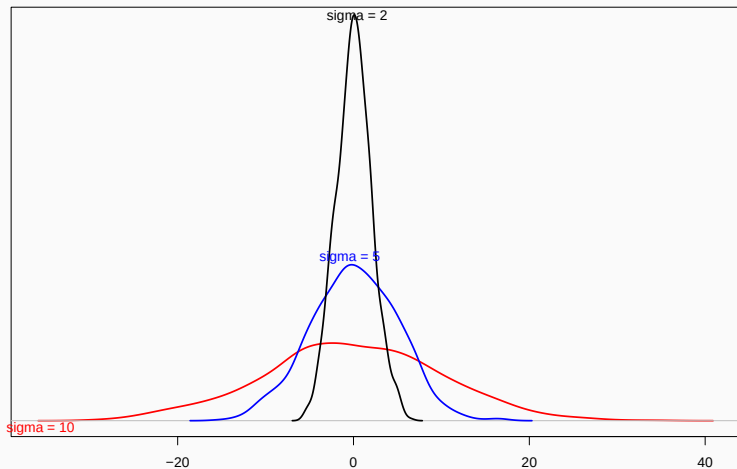
- points farther from regression line
- larger difference of observed - predicted values



Residual variation (sigma) is the Std. Dev. of residuals

$$\varepsilon_i \sim N(0, \sigma^2)$$

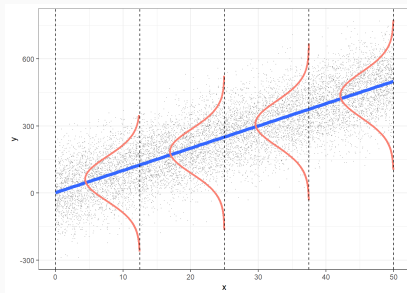
Distribution of residuals



In a general linear model we assume residuals are

$$\varepsilon_i \sim N(0, \sigma^2)$$

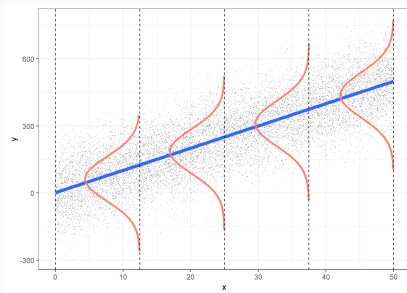
- Normal



In a general linear model we assume residuals are

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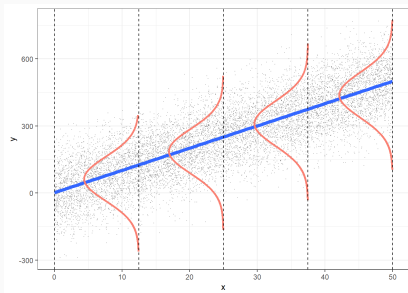
- Normal
- Centred on 0 (no bias)



In a general linear model we assume residuals are

$$\varepsilon_i \sim N(0, \sigma^2)$$

- Normal
- Centred on 0 (no bias)
- Homogeneous variance (*homoscedasticity*)



Different ways to write same model

$$y_i = a + bx_i + \varepsilon_i$$
$$\varepsilon_i \sim N(0, \sigma^2)$$

$$y_i \sim N(\mu_i, \sigma^2)$$
$$\mu_i = a + bx_i$$
$$\varepsilon_i \sim N(0, \sigma^2)$$

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